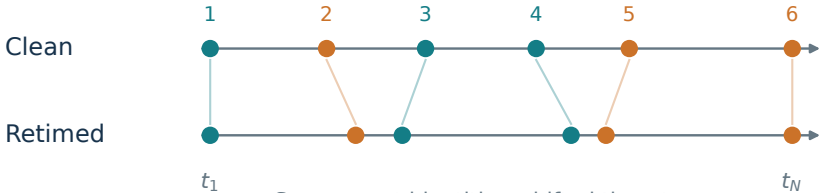
a Retiming changes when events arrive



Same event identities; shifted timestamps

Preserved

Coordinates and polarity
Event identities and count
Labels and recording window

Intervals and bin counts can change.

b Two separate experimental tracks

TIMESTAMP

Native events → 10 equal-duration bins

$|t_i' - t_i| \le \varepsilon_f D;\ \ \varepsilon_f = 5\%,\ 10\%$

Order retained; equal times allowed

QSNN-v3 + compact SNN · 100 validation examples

DISCRETE GRID

Frozen 10-bin equal-event-count grid

$B_\infty = 1$ bin or $B_0 = 200$ packets

Packet amplitude retained; no collisions

Four classical models · 1,000 test examples / repr.

c QSNN-v3: simulated hybrid classifier

Conv / LIF
16 → 32 channels
spikes + membrane

32D latent
32 → 8 projection
8 → 8 angle map

8 qubits · 2 blocks
Upload · Train · CNOT ×2

256
probabilities
learned basis

10
classes
linear

Analytic state-vector simulation · 40,762 trainable parameters